Channel1

STEP1 STEP

DIR1 DIR

File: Channel.kicad_sch

Channel2

STEP2 STEP

DIR2 DIR

File: Channel.kicad_sch

Channel3

STEP3 STEP

DIR3 DIR

File: Channel.kicad_sch

Channel4

STEP4 STEP

DIR4 DIR

File: Channel.kicad_sch

Channel5

STEP5 STEP

DIR5 DIR

File: Channel.kicad_sch

Channel6

STEP6 STEP

DIR6 DIR

File: Channel.kicad_sch

Channel7

STEP7 STEP

DIR7 DIR

File: Channel.kicad_sch

Channel8

STEP8 STEP

DIR8 DIR

File: Channel.kicad_sch

Channel9

STEP9 STEP

DIR9 DIR

File: Channel.kicad_sch

Channel10

STEP10 STEP

DIR10 DIR

File: Channel.kicad_sch

Channel11

STEP11 STEP

DIR11 DIR

File: Channel.kicad_sch

Channel12

STEP12 STEP

DIR12 DIR

File: Channel.kicad_sch

Control

File: Control.kicad_sch

Power

File: Power.kicad_sch

Sheet: /
File: 12StepperPCB.kicad_sch

Title: 12-Channel Stepper Driver Carrier

Size: A2	Date:	**Rev:**
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VMOT: 30V max, 12A bus (1A/ch target). Dual entry, second terminal unfused —
for low-resistance current injection only, not a redundant protection path.
A4988 max 35V; TMC2209 max 28V operating (29V abs max). Above 28V: A4988-only.
Fuse footprint left TBD — confirm blade-fuse holder or polyfuse P/N before layout.

J51



VDD_5V_IN

J52



VMOT_IN_A

J53



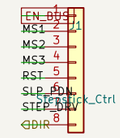
VMOT_IN_B

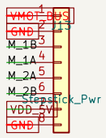


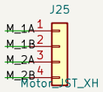


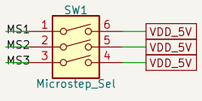


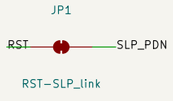
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KiCad E.D.A. 10.0.5		Id: 2/15

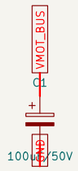


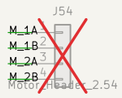




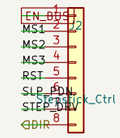


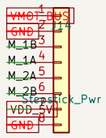


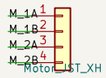


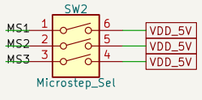


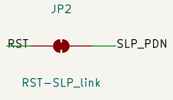
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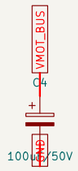


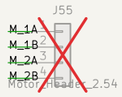




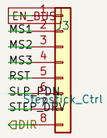


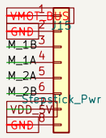


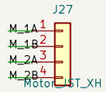


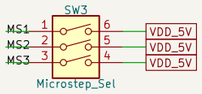


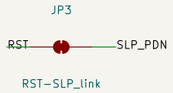
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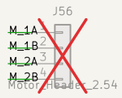




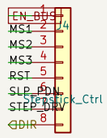


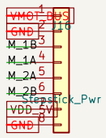


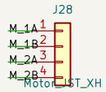


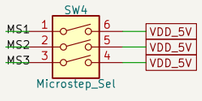


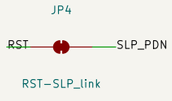
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KiCad E.D.A. 10.0.5		Id: 5/15



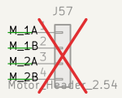




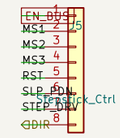


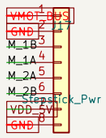


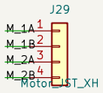


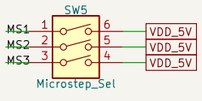


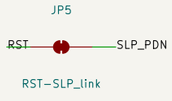
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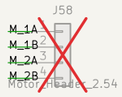




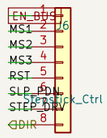


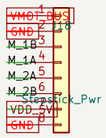


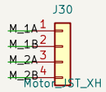


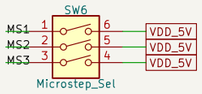


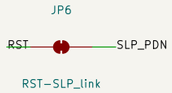
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KiCad E.D.A. 10.0.5		Id: 7/15



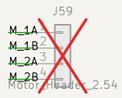




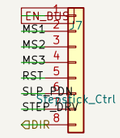


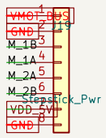


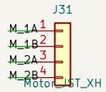


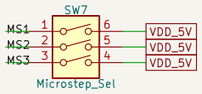


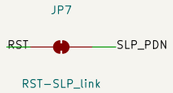
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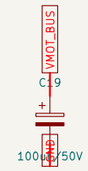


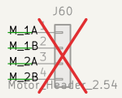




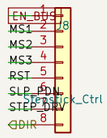


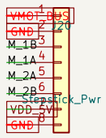


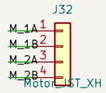


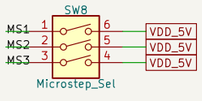


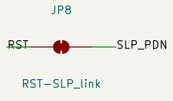
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KiCad E.D.A. 10.0.5		Id: 9/15

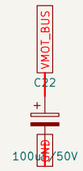


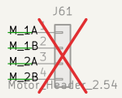




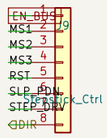


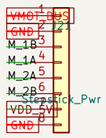


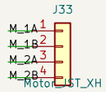


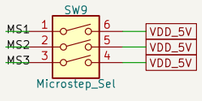


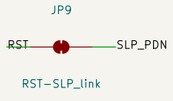
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KiCad E.D.A. 10.0.5		Id: 10/15



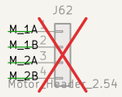




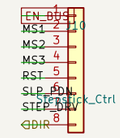


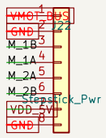


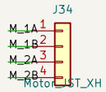


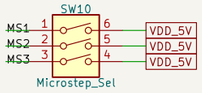


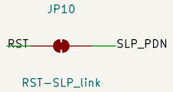
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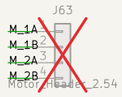




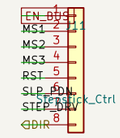


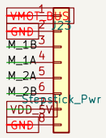


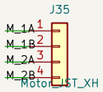


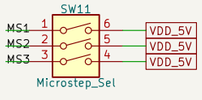


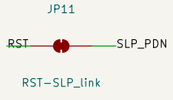
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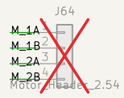




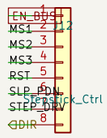


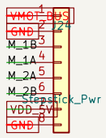


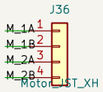


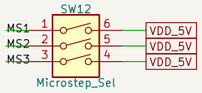


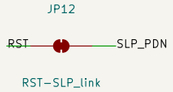
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KiCad E.D.A. 10.0.5		Id: 13/15



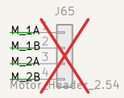








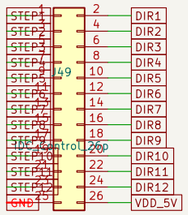




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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 14/15

EN header: center=EN_BUS, pin1=GND (jumper 1–2 = always enabled),
pin3=EN_MCU breakout for MCU-driven global enable (external jumper to center).





el): odd pins 1,3..23=STEP1–12, even pins 2,4..24=DIR1–12, 25=GND, 26=VDD (5V sense/supply). Each channel's STEP/DIR pair shares one connector column and adjacent ribbon wires.

Sheet: /Control/		
File: Control.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 15/15